LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 10 Test: The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Draw neat, labelled ray diagrams wherever required.

SECTION A

6 marks

Choose the correct option.

Q1. The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to: **[1]**

(a) presbyopia (b) accommodation
(c) near-sightedness (d) far-sightedness

Q2. The human eye forms the image of an object at its: **[1]**

(a) cornea (b) iris
(c) pupil (d) retina

Q3. The least distance of distinct vision for a young adult with normal vision is about: **[1]**

(a) 25 m (b) 2.5 cm
(c) 25 cm (d) 2.5 m

Q4. The change in focal length of the eye lens is caused by the action of the: **[1]**

(a) pupil (b) retina
(c) ciliary muscles (d) iris

Q5. Myopia (near-sightedness) is corrected using a: **[1]**

(a) concave lens (b) convex lens
(c) bifocal lens (d) cylindrical lens

Q6. **Assertion (A):** The sky appears blue on a clear day.
Reason (R): Blue light has a shorter wavelength and is scattered much more strongly by the fine particles in the atmosphere than red light. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why do stars twinkle? **[2]**

OR

Why does the Sun appear reddish at sunrise and sunset?

Q8. What is dispersion of white light? What causes it? **[2]**

SECTION C

6 marks

Q9. A student can see distant objects clearly but cannot read a book held at a normal distance. **[3]**

(i) Name this defect of vision. **[1]**

(ii) Give two possible causes. **[1]**

(iii) Name the lens used to correct it, and say how it helps. **[1]**

Q10. The Sun is visible about 2 minutes before actual sunrise and about 2 minutes after actual sunset. Explain why. **[3]**

OR

What is the Tyndall effect? Why are danger signal lights red in colour?

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What is myopia? Give two causes, and explain how it is corrected. **[3]**

(ii) What is presbyopia? How is it corrected? **[2]**

OR

(a) Describe the formation of a spectrum by a glass prism. Which colour bends the least and which the most? [3]
(b) How did Newton show that white light is made of seven colours? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A rainbow is a natural spectrum that appears in the sky after a rain shower. It is caused by the dispersion of sunlight by tiny water droplets in the atmosphere. A rainbow always forms in the direction opposite to the Sun. The droplets act like small prisms: they refract and disperse the sunlight, reflect it internally, and refract it again as it comes out of the droplet. **[4]**

(i) In which direction must you face to see a rainbow? **[1]**

(ii) What acts as a prism in the formation of a rainbow? **[1]**

(iii) Name, in order, the three phenomena of light that occur inside a raindrop. **[2]**

~ All the best! ~